

KUSHAGRA MISHRA

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SUMMARY

AI Engineering Intern at Tech Mahindra (Makers Lab) with hands-on experience building and deploying production agentic AI systems using LangGraph, RAG pipelines, ChromaDB vector search, FastAPI, and React. B.Tech in Computer Science (2026) with strong Java fundamentals and from-scratch ML experience in PyTorch. Seeking AI/ML Engineer or GenAI Developer roles.

PROFESSIONAL EXPERIENCE

AI Engineering Intern — Tech Mahindra, Makers Lab, Hyderabad Jun 2026 – Present

- Built an in-app AI guidance chatbot — a 4-node LangGraph pipeline (intent routing, knowledge retrieval, capability matching, response generation) with FastAPI backend and React frontend
- Implemented semantic search using ChromaDB vector embeddings (all-MiniLM-L6-v2), replacing full-context injection with relevance-based retrieval for both knowledge base and capability matching
- Optimized pipeline from 3 to 2 LLM calls per query (~33% fewer calls); deployed full stack to production (Render + ChromaDB Cloud + Netlify) with a documented GCP migration path (Cloud Run, Firebase Hosting, Secret Manager)
- Redesigned session management with server-side routing and session-scoped storage; authored technical reports on agentic architectures, vector databases, semantic search, and GCP deployment

PROJECTS

MicroGPT + Attention X-Ray *Python, PyTorch, React*

- Implemented a decoder-only transformer from scratch in PyTorch — multi-head self-attention, positional encodings, layer normalization, and full training loop without high-level libraries
- Built a browser-based attention visualization UI rendering weights across every head and layer for model interpretability

Enterprise AI Workspace *React, FastAPI, MySQL, ChromaDB, LangChain*

- Full-stack AI document workspace with RAG pipeline (ChromaDB vector search) for contextual Q&A, semantic search, and summarization; scalable REST APIs integrating LLM workflows via LangChain

MediaVault (In Progress) *Java 17, Spring Boot, Spring Security, JPA, MySQL, React*

- Media tracking platform with layered Spring Boot REST API, JWT auth (BCrypt + custom filter chain), and an LLM-powered recommendation endpoint generating personalized suggestions from user history

SKILLS

Languages: Java, Python, JavaScript, SQL

AI & GenAI: LangChain, LangGraph, CrewAI, RAG, Prompt Engineering, LLM Integration, PyTorch, Transformers

Vector & ML: ChromaDB, Vector Embeddings, Semantic Search, Sentence Transformers, Scikit-learn, Pandas, NumPy

Backend & Web: Spring Boot, FastAPI, REST APIs, React.js, HTML5, CSS3

DevOps & Cloud: Git, GitHub, Docker, Render, Netlify, GCP (Cloud Run, Firebase Hosting, Secret Manager), Postman

EDUCATION

B.Tech, Computer Science and Engineering — Vellore Institute of Technology Sep 2022 – May 2026 | CGPA: 8.19

Class XII (CBSE) — Jawahar Vidya Mandir, Ranchi 2022 | 89%

Class X — St. Thomas School, Ranchi 2020 | 96%